Solar Energy Basics

Solar panels convert sunlight into electricity using photovoltaic cells made mostly of silicon.

A typical residential solar panel has an efficiency between 15 and 22 percent.

Most manufacturers guarantee that panels will still produce at least 80 percent of their rated power after 25 years.

Costs and Maintenance

In 2023 the average installed cost of a home solar system was about 2.8 dollars per watt.

Panels need very little maintenance; cleaning them twice a year is usually enough.

Inverters typically need to be replaced after 10 to 15 years.